Experiment M20bis
Motion in rotating reference frames

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Abstract

This laboratory experiment consisted of analyzing the uniform rectilinear motion of a sphere from the perspective of a rotating, non-inertial reference frame. The primary goal was to visually capture and quantitatively analyze how this simple motion appears as a complex, curved trajectory when observed from a rotating system. Furthermore, the study aimed to determine the angular velocity of the frame through two independent methods: direct measurement with a photogate and indirect calculation from the kinematic data of the recorded motion.

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1 Introduction

The analysis of motion is fundamentally dependent on the observer's frame of reference. While inertial frames provide a simple description of motion via Newton's laws, many real world systems, such as planetary atmospheres or navigation systems, are best analyzed from rotating, non-inertial perspectives.

This laboratory practice investigates the principles of motion as observed from a uniformly rotating reference frame. The main objective is to experimentally analyze the uniform rectilinear motion of a sphere from the viewpoint of a rotating camera. While the sphere moves in a straight line at constant speed in the laboratory (inertial) frame, its trajectory appears as a complex, looped curve in the rotating system due to the apparent Coriolis (Figure 1) and centrifugal accelerations.

By recording the motion under different rotational speeds, we will quantify these effects and validate the theoretical equations for rotating frames, highlighting the strong influence of the reference frame on the description of motion.



Figure 1: Hurricane produced by the Coriolis effect of the atmosphere.

1.1 Theorical principles

According to Newton's First Law, an object subjected to a net force of zero will move with a constant velocity, describing a straight-line path in an inertial reference frame (I.R.F.). Assuming motion in the xy-plane, the coordinates of such an object are given by:

$$x(t) = x_0 + v_{0x}t \tag{1}$$
$$y(t) = y_0 + v_{0y}t$$

However, the description of this same motion changes drastically when observed from a rotating, non-inertial reference frame (N.I.R.F.). From this perspective, the simple rectilinear path appears as a curved, often looped, trajectory. The shape and complexity of this apparent path, as seen in Figure 2, depend critically on the ratio between the rotational speed of the frame (ω) and the absolute speed of the object (v_0). A higher ω/v_0 ratio can even result in trajectories with multiple loops.

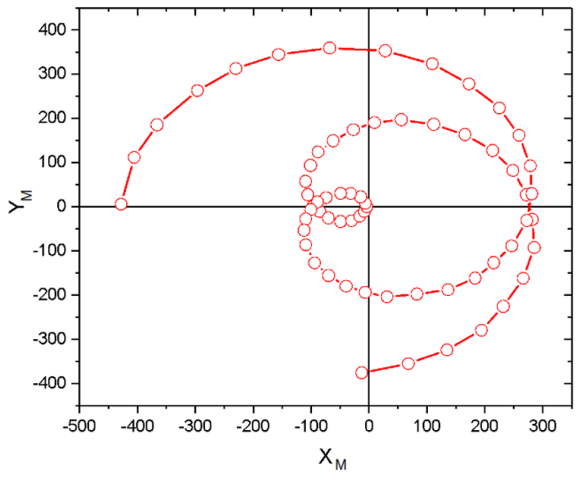


Figure 2: Trajectory of the mobile seen from the rotating reference frame at maximum engine speed [1].

The acceleration of a particle measured in a N.I.R.F. rotating with a constant angular velocity ω about a fixed axis is given by

$$\vec{a}_m = -\vec{\omega} \times (\vec{\omega} \times \vec{r}_m) - 2(\vec{\omega} \times \vec{v}_m), \tag{2}$$

where $\vec{r}_m$ and $\vec{v}_m$ are the position and velocity vectors in the N.I.R.F., respectively. The term $-\vec{\omega} \times (\vec{\omega} \times \vec{r}_m)$ is the centrifugal acceleration, and $2(\vec{\omega} \times \vec{v}_m$ is the Coriolis acceleration.

Assuming the axis of rotation is perpendicular to the plane of motion, this vector equation can be expressed in Cartesian components as the following system of coupled second-order differential equations:

$$\begin{cases} a_{M,x} = \omega^2 x_M + 2\omega v_{M,y} \\ a_{M,y} = \omega^2 y_M + 2\omega v_{M,x} \end{cases} \tag{3}$$

To facilitate experimental verification, these equations can be recombined into forms that are more suitable for linear regression analysis. Multiplying and combining the components leads to:

$$\begin{cases} a_{M,x} v_{M,x} + a_{M,y} v_{M,y} = \omega^2 (x_M v_{M,x} + y_M v_{M,y}) \\ a_{M,x} y_M - a_{M,y} x_M = 2\omega (v_{M,x} x_M + v_{M,y} y_M) \end{cases} \tag{4}$$

The second equation, in particular, provides a direct method to determine the angular velocity ω from measured kinematic data.

Finally, to recover the original straight-line trajectory as seen from the I.R.F., the coordinates measured in the N.I.R.F. must be transformed. For a frame rotating counterclockwise, this transformation is achieved by applying the rotation matrix:

$$\begin{pmatrix} cos(\omega t) & -sin(\omega t) & 0 \\ sin(\omega t) & cos(\omega t) & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} x_M \\ y_M \\ 0 \end{pmatrix} = \begin{pmatrix} x \\ y \\ 0 \end{pmatrix} \tag{5}$$

where it is assumed that the two frames coincide at time $t = 0$. In practice, an initial phase angle must be included to account for the orientation of the N.I.R.F. at the start of the recording.

2 Materials and methods

The information for this section is provided in the Laboratory Manual [1].

2.1 Materials

In this experiment we used the following materials.

- **Wi-Fi camera**, which was used as a reference system, as well as to record all the experiences that will be analysed later. It could record up to 120 frames per second.
- **Motor**. This instrument enabled us to rotate the camera, and with that we could construct an S.R. in constant rotation..
- **Power supply**, used to put the motor in motion.
- **Photogate**: it is a device that projects and receives a laser in order to measure the speed of the camera, which has a collar to interrupt the optical path of the laser. It has a sensitivity of 1 ms.
- **Marble sphere**, thrown in the experiment.
- **Ramp and straight rail**, from where the sphere is thrown.

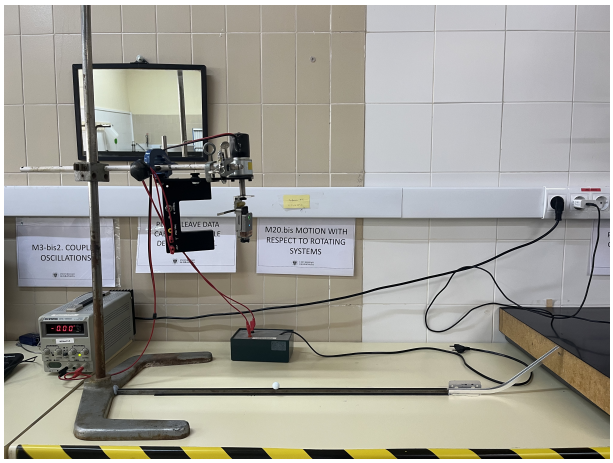


Figure 3: Laboratory setup for the experiment on motion in rotating systems.

2.2 Methods

2.2.1 Presets

First of all, we need to make sure the rail is situated correctly, between the marks made on the table. Now we will connect our phone to the camera, which must be static. Once we have achieved a stable connection, we are going to record the sphere at rest with the camera turning at its full velocity. This will be helpful when positioning the R.F. later.

2.2.2 Uniform rectilinear movement

This section must be repeated thrice, in order to obtain a better result. We will record, with the camera at rest, the trajectory of the sphere after positioning it at the highest point of the ramp. It is important to not apply an initial velocity. When we analyse these videos we must avoid the peripheral distorsion of the camera (its lens is not flat).

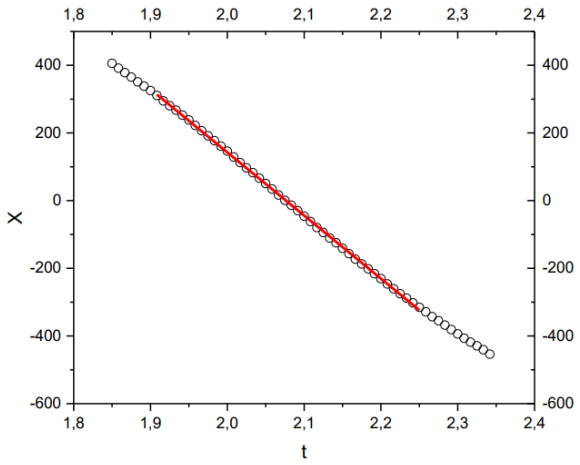


Figure 4: Peripheral distorsion of the camera[1].

2.2.3 Angular velocity ω

Firstly, we are going to apply a voltage to the motor and measure the period with the photogate. We can obtain the angular velocity ω with the relation:

$$\omega = \frac{2\pi}{T} \left(\frac{rad}{s}\right). \tag{6}$$

Once again we need to repeat this experience three times and calculate the average value of ω.

2.2.4 Trajectory observed from the non-inertial R.F.

With the same voltage used in the previous section, we must rotate the camera and launch the sphere. The recording will show how the trajectory of the mass is seen from the camera in rotation. This and the previous experiment need to be repeated for voltages 11, 10 and 9 volts.

2.2.5 Video analysis

For each recording we must complete the following objectives using the program Tracker 6.

- Cut the video in order to show only the stills when the sphere appears.
- Define the calibration rod on the width of the rail, in the centre of the image. This allows the program to assign a pixel to a number of centimeters or meters in real life.
- Track the point mass (our marble sphere) automatically. To do so, the evaluation rate must be at its 5%, the tether at 0% and the automark at 1.
- Define S.R. over the turning axis and the x_m axis, following the direction and orientation of the initial velocity of the sphere. Figure 2 can be helpful.
- Save and organise the data from the videos (x, y, v_x, v_y, a_x and a_y).
- Using the recordings from the previous experiment, obtain the initial angle between the axis x (when the camera is not rotating) with respect to x_M. This is the lag that must be included in transformation 5.

3 Results and discussion

3.1 Linear velocity v_0 of the sphere

In this section we must prove that the sphere describes an uniform rectilinear movement, but this is not true due to the friction force. However, considering the trajectory as an U.R.M. is a very good approximation because the contact surface is quite small, as well as the length of the rail.

To verify the goodness of this approximation, we can study the speed of the sphere. After analysing the three videos, we obtained the following velocities.

	Video 1	Video 2	Video 3
$v_0(m/s)$	**0,545 ± 0,010**	**0,5617 ± 0,0070**	**0,5587 ± 0,0052**

Table 1: Approximate values of the sphere's velocity with its corresponding uncertainty for each video.

When analysing the videos with Tracker 6 we observed that the v *vs.* t graphics described a parabola with negative slope instead of a constant function. We attribute this error to the peripheral distorsion of the camera and the friction force. The data collected in Table 1 allows us to provide an average value of v_0.

$$\overline{v_0} = 0,5551 \pm 0,0045 \ (m/s).$$

3.2 Calculation of ω

The angular velocity (ω) of the rotating camera, which defines the Non Inertial Reference Frame (N.I.R.F.), was determined using two independent methods:

1. **Direct Measurement using Photogate**: using a photogate system, the period of rotation (T) was measured three times for each supply voltage, and the angular velocity was calculated using equation 6. The uncertainty was propagated from the period measurements using standard error propagation formulas.

2. **Indirect Calculation from Kinematic Data in N.I.R.F.**: an alternative method to determine ω ca be derived by theoretical relationship derived from the equations of motion in rotating frames (4). The second equation of this system represents a linear relationship of the form $Y = 2\omega X$, where

$$\begin{cases} Y = a_{M,x} y_M - a_{M,y} x_M \\ X = v_{M,x} x_M + v_{M,y} y_M \end{cases} \tag{7}$$

   The position, velocity, and acceleration components in the N.I.R.F. were obtained through video analysis using Tracker software. Then, a linear regression was performed on the (X, Y) data pairs, and the angular velocity was calculated from the slope (a) as:

$$\omega_2 = \frac{a}{2} \tag{8}$$

   The uncertainty of ω_2 was derived from the standard error of the slope provided by the regression analysis.

The results from both methods for the different supply voltages are summarized in Table 2.

Voltage (V)	**T (s)**	ω_1 **(rads/s)**	ω_2 **(rads/s)**	ε
11.00 ± 0.01	0.4322 ± 0.0018	-14.5388 ± 0.0593	-18.5531 ± 1.7181	27.61 %
10.00 ± 0.01	0.4750 ± 0.0008	-13.2278 ± 0.0213	-14.5506 ± 2.5260	10.00%
9.00 ± 0.01	0.5335 ± 0.0021	-11.7773 ± 0.0460	-11.9272 ± 1.0572	1.27%

Table 2: Comparison of angular velocities obtained via photogate measurement (ω_1) and kinematic analysis (ω_2), with its corresponding uncertainties and relative errors ε.

The results demonstrate a clear correlation between the supply voltage and rotational speed, with higher voltages producing shorter periods and, as a consequence, higher angular velocities. The agreement between both methods improves significantly at lower voltages: relative error decreases strongly from 27.61% to 1.27%. This change suggests that experimental conditions at lower rotational speeds provide more reliable kinematic data, likely due to reduced motion blur and improved tracking accuracy.

3.3 Comparison of the trajectories observed from the non-inertial and inertial S.R.

Finally, the trajectory of the sphere was reconstructed for a camera supply voltage of $10,01 \pm 0,01V$. To obtain the positions of the sphere in the I.R.F. we used transformation 5. The initial angle between the rail and the axis x_M+ we measured was $\theta_0 = \frac{\pi}{50}\ rad$, with a temporal resolution of $\Delta t = \frac{1}{120}\ s$. The angular velocity of the rotating camera was determined to be $\omega = -13,228 \pm 0,021\ \frac{rad}{s}$ (it rotates clockwise).

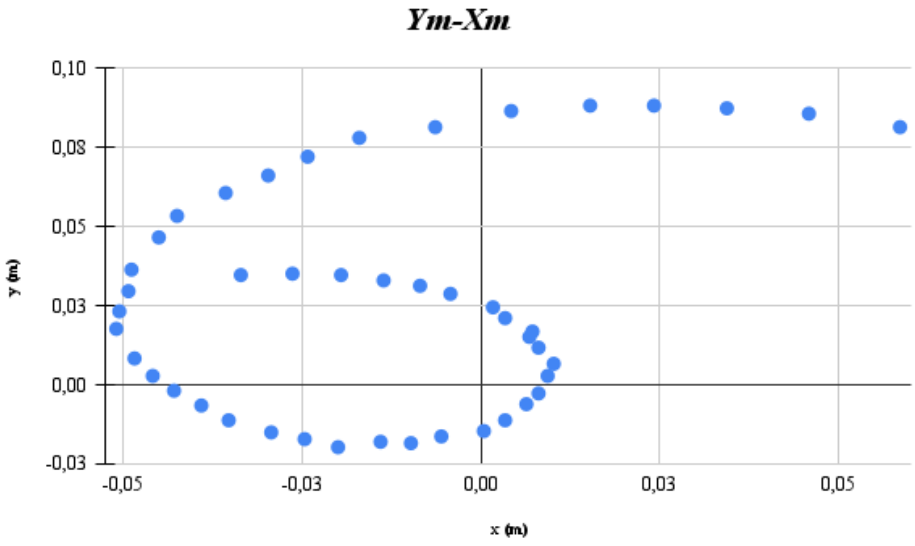


Figure 5: Trajectory of the sphere observed from the N.I.R.F.

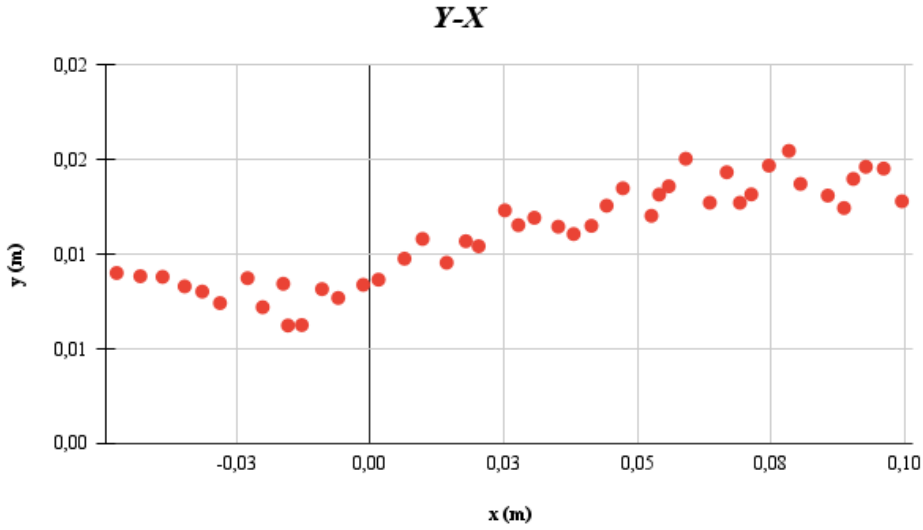


Figure 6: Trajectory of the sphere observed from the I.R.F.

The first graph depicts a single frame from the full trajectory presented in Figure 2. The camera's limited field of view prevented the capture of the complete trajectory within a single image. In contrast, Figure 6 is expected to exhibit a linear relationship with a positive slope. The experimental data does suggest a linear trend, however, the deviation from an ideal linear fit is significant. This discrepancy is likely attributable to the camera's peripheral distortion and errors in the measured values of ω and θ_0.

An important consideration is the apparent offset of the fitted functions from the Y-axis. This translational discrepancy suggests a misalignment in the experimental setup, where the rail was positioned above the horizontal axis, probably due to camera movement.

4 Conclusion

In conclusion, this experiment successfully showed how the same trajectory looks completely different depending on whether it is observed from an inertial or a non-inertial reference system. The uniform rectilinear motion of the sphere in the I.R.F. was seen as a curved, looped path from the rotating N.I.R.F., providing a visual of the Coriolis effects.

We calculated the angular velocity (ω) in two different ways. The results from the photogate and the video analysis matched much better at lower speeds, which reveals that it is easier to get accurate measurements when the camera rotates slower.

Finally, we used coordinate transformation 5 allowed for the reconstruction of the original inertial trajectory from the data collected in the rotating frame. While the resulting plot showed a linear tendency, the deviations from an ideal straight line underline the influence of systematic errors, including peripheral camera distortion and miscalculations of the initial angle θ_0 and ω.

5 Appendixes

5.1 General Uncertainties Calculations

As established in measurement uncertainty analysis, Type A uncertainty, denoted as $u_A(x)$, quantifies the statistical variation associated with repeated observations and is calculated using the equation:

$$u_A(x) = \frac{\sigma(x)}{\sqrt{N}}, \tag{9}$$

where $\sigma(x)$ represents the standard deviation and N the number of observations. On the other hand, Type B uncertainty, $u_B(x)$, is derived from the estimated accuracy of the measuring instrument and is obtained from:

$$u_B(x) = \frac{\delta_x}{\sqrt{12}}, \tag{10}$$

denotes the instrument's resolution ($\delta_V = 0.01$ V for voltage measurements and $\delta_T = 0.001$ s for the photogate period measurements). The combined standard uncertainty, $u_C(x)$, is subsequently expressed as the root sum of squares of these components:

$$u_C(x) = \sqrt{u_A^2(x) + u_B^2(x)}. \tag{11}$$

For derived quantities obtained through functional relationships $y = f(x_1, x_2, \ldots, x_n)$, the uncertainty was propagated using:

$$u_C(y) = \sqrt{\sum_{i=1}^{n} \left(\frac{\partial f}{\partial x_i}\right)^2 u_C(x_i)^2}. \tag{12}$$

These formulas were extracted from the document [2].

5.2 Specific Uncertainty Calculations

5.2.1 Period Measurement (T)

The rotational period was measured three times for each voltage setting. The combined uncertainty was calculated as:

$$u_C(T) = \sqrt{u_A(T)^2 + u_B(T)^2}, \tag{13}$$

where:

- $u_A(T) = \dfrac{\sigma(T)}{\sqrt{3}}$ (from three period measurements)
- $u_B(T) = \dfrac{0.001}{\sqrt{12}}$ (photogate resolution of 0.001s)

5.2.2 Linear Velocity v_0

The sphere's linear velocity for each video was determined by calculating the mean of the dataset extracted from Tracker 6 software, which was assigned a tolerance of 7%. Consequently, its corresponding combined uncertainty is given by:

$$u_C(v_0) = \frac{1}{N}\sqrt{u_C(v_1)^2 + u_C(v_2)^2 + ... + u_C(v_N)^2}\ (m/s),$$

where N represents the total number of data points collected. The average value $\overline{v_0}$ of these three resultant velocities was calculated analogously.

$$u_C(\overline{v_0}) = \frac{1}{3}\sqrt{u_C(v_{01})^2 + u_C(v_{02})^2 + u_C(v_{03})^2}\ (m/s).$$

5.2.3 Angular Velocity from Photogate (ω_1)

For the angular velocity obtained from the photogate period measurements we obtain, after applying equation 12, the following relation:

$$u_C(\omega_1) = \left|\frac{\partial \omega_1}{\partial T}\right| u_C(T) = \frac{2\pi}{T^2} u_C(T). \tag{14}$$

5.2.4 Angular Velocity from Kinematic Analysis (ω_2)

The uncertainty in ω_2 obtained from linear regression of the kinematic data corresponds to the standard error of the slope:

$$u_C(\omega_2) = \frac{\mathrm{b}}{2}, \tag{15}$$

where b is the standard error of the slope provided by the linear regression analysis.

5.2.5 Relative Error

The relative error between the two methods was calculated as:

$$\varepsilon_r = \left|\frac{\omega_2 - \omega_1}{\omega_1}\right| 100\%. \tag{16}$$

5.3 Graphs referring to linear adjustments

This section shows the different linear adjustment graphics used to obtain ω_2 for each voltage.

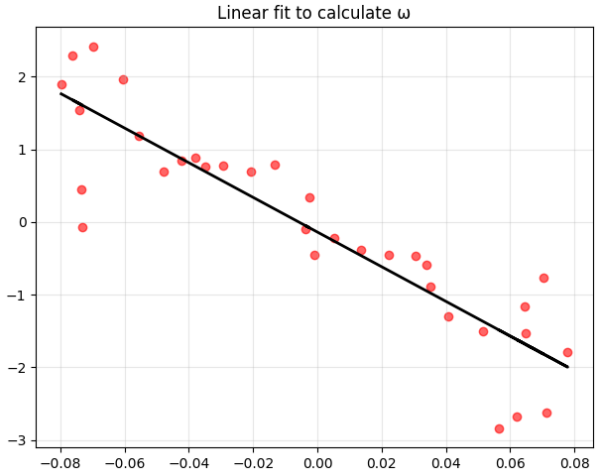


Figure 7: Linear adjustment for a $9.00 \pm 0.01V$ voltage.

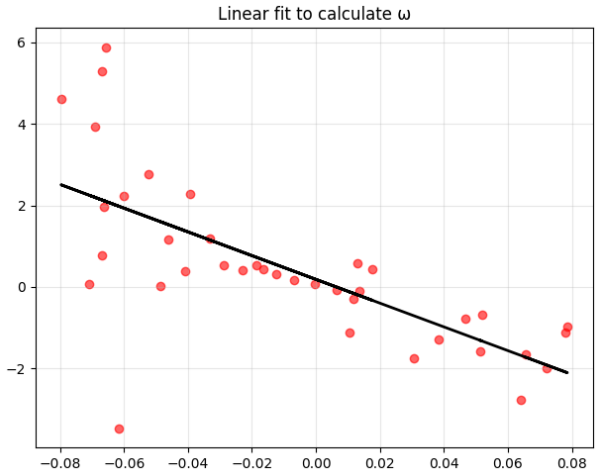


Figure 8: Linear adjustment for a $10.00 \pm 0.01V$ voltage.

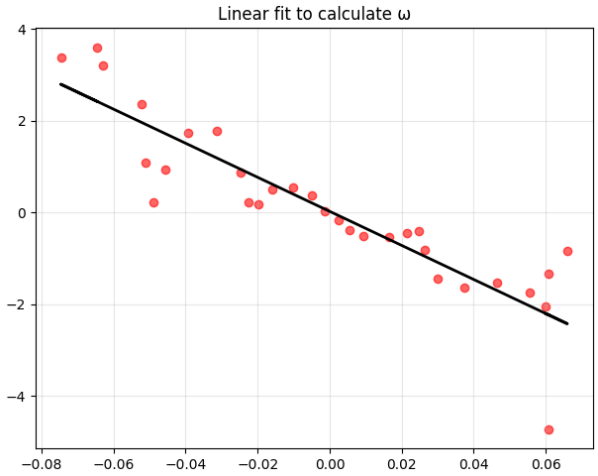


Figure 9: Linear adjustment for a $11.00 \pm 0.01V$ voltage.

References

[1] Miguel Ángel Rodríguez Valverde; Artur Schmitt; José Callejas Fernández; Miguel Cabrerizo Vílchez. *MOVIMIENTO DESDE SISTEMAS EN ROTACIÓN*. Department of Applied Physics, University of Granada, 2023.

[2] Department of Applied Physics; University of Granada. *Notes for the Calculation of Uncertainties in Laboratory Sessions*. PRADO, 2025.